Assignment 2 - Git & GitHub Fundamentals Report Template

• Group Size: 2 students

• Chosen Theme: Quiz

• Full Names: Rayan Benlachhab - Mohamed Amine Menouer

• Student IDs: 158274 - 145847

• Course Codes & Sections: CSC332401

• GitHub Usernames: benlachhabrayan-glitch / penouar

• GitHub Repository URL: https://github.com/benlachhabrayan-glitch/Github-assignment-2

• GitHub Pages URL: https://benlachhabrayan-glitch.github.io/Github-assignment-2/

1. Project & Role Distribution Briefly describe the chosen theme, the feature areas, and who was responsible for each one. State whether the group followed the 2-member or 3-member path.

Our group followed the 2-member project path and chose a quiz project as our theme. The project allows users to answer six questions by selecting the number corresponding to their chosen answer from the displayed options. Once all questions have been answered, the program calculates the total number of correct answers and displays the final score as a percentage of correctness.

Rayan Benlachhab was responsible for the entire code implementation, including the quiz functionality, answer validation, score calculation, and unit tests. Mohamed Amine Menouer was responsible for collaborating on the project through the required GitHub collaboration workflow, including code review and pull request activities.

2. Branches, Commits & Contributions For each member, identify the main feature branch, summarize the work completed, and mention the meaningful commits made.

Rayan worked on the feature/quiz-improvements, he developed the logic behind the quiz, the display of the score, the percentage of good answers, 3-unit tests, the settings of the

repository, the rulesets and the activation of the workflows of GitHub and its checking tests.

```
PROBOOK@DESKTOP-7L7SVS9 UCRT64 ~/Downloads/Github assignment 2 practice (feature/pages-final-update)
$ git branch -a
  a
  demo/check-required
  demo/review-required
* feature/pages-final-update
  feature/project-page
  feature/quiz-improvements
  main
```

```
  * feature/pages-final-update
    feature/project-page
    feature/quiz-improvements
    main
    remotes/origin/HEAD -> origin/main
    remotes/origin/benlachhabrayan-glitch-patch-1
  :
```

```
PROBOOK@DESKTOP-7L7SVS9 UCRT64 ~/Downloads/Github assignment 2 practice (feature/pages-final-update)
$ git log --oneline --all
59fdca6 (HEAD -> feature/pages-final-update, origin/feature/pages-final-update, a) modification web site
76a48e7 (origin/main, origin/HEAD, main) Merge pull request #8 from benlachhabrayan-glitch/feature/project-page
ada69c9 (origin/feature/project-page, feature/project-page) Add static project page
e06d258 (origin/demo/check-required, demo/check-required) trying to test the invalidation of the check
b08b8a7 (origin/demo/review-required, demo/review-required) trying to check the automated test
1fa9f86 Merge pull request #5 from benlachhabrayan-glitch/feature/quiz-improvements
5e00155 (origin/feature/quiz-improvements, feature/quiz-improvements) resolve conflict
```

```
e06d258 (origin/demo/check-required, demo/check-required) trying to test the invalidation of the check
b08b8a7 (origin/demo/review-required, demo/review-required) trying to check the automated test
1fa9f86 Merge pull request #5 from benlachhabrayan-glitch/feature/quiz-improvements
5e00155 (origin/feature/quiz-improvements, feature/quiz-improvements) resolve conflict
2a3e881 Merge pull request #4 from benlachhabrayan-glitch/feature/readme-status-amine
3bbb1fd update to README
:
```

```
PROBOOK@DESKTOP-7L7SVS9 UCRT64 ~/Downloads/Github assignment 2 practice (feature/pages-final-update)
$ git log --all --format="%h | %an | %s"
59fdca6 | Ray | modification web site
76a48e7 | Mohamed Amine Menouer | Merge pull request #8 from benlachhabrayan-glitch/feature/project-page
ada69c9 | Ray | Add static project page
e06d258 | Ray | trying to test the invalidation of the check
b08b8a7 | Ray | trying to check the automated test
1fa9f86 | Mohamed Amine Menouer | Merge pull request #5 from benlachhabrayan-glitch/feature/quiz-improvements
5e00155 | Ray | resolve conflict
```

```
1fa9f86 | Mohamed Amine Menouer | Merge pull request #5 from benlachhabrayan-glitch/feature/quiz-improvements
5e00155 | Ray | resolve conflict
2a3e881 | benlachhabrayan-glitch | Merge pull request #4 from benlachhabrayan-glitch/feature/readme-status-amine
3bbb1fd | Ray | update to README
:
```

Mohamed Amine worked on feature/quiz-engine, added 3-unit tests including a zero-

questions edge-case test after review feedback, and participated in pull request reviews and README conflict resolution.

3. Pull Requests & Code Review Identify each member's Pull Request and review responsibility. Describe the Request changes cycle and how the author updated the same branch before approval.

Rayan Benlachhab made a pull request after committing an improvement for the quiz logic and 3-unit tests with the addition of the GitHub actions test workflow. His pull request was accepted directly after a small review from Amine.

Amine made a pull request after committing an addition of 2-unit tests into the project, but he received from Rayan a request changes review to add modification by adding another unit test for handling edge case scenario. He updated the code and tests on the same feature branch, then pushed the changes to the existing Pull Request, and after that, the pull request was approved after a small review from Rayan.

4. Merge Conflict Explain how the required README conflict was created, what Current and Incoming represented in your case, and how the group decided the final resolved content.

To demonstrate conflict resolution, we modified the same project-status line in the README file on separate branches. Since the changes affected the same line, Git identified a merge of conflict after accepting one PR and another PR still open. So, we started by fetching the latest remote changes and merging origin/main into the feature branch.

Current represented the feature branch version, while Incoming represented the changes from main. Git marked the conflicting section, allowing us to compare the current and incoming versions. We resolved the conflict by both deleting conflicted updates and making a new one, committing the solution, and pushing the updated branch.

Finally, the second pull request was reviewed and merged. This exercise demonstrated how Git handles conflicting edits and how teams can resolve them before integrating

changes into the main branch.

```
PROBOOK@DESKTOP-7L7SVS9 UCRT64 ~/Downloads/Github assignment 2 practice (feature/quiz-improvements)
$ git merge origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
```

5. Unit Tests Summarize what your tests verify. Identify examples of normal behavior, an edge/boundary case, and invalid input/error handling.

Our project includes a total of six-unit tests. Two of them test normal behavior by checking whether the program correctly handles correct and incorrect answers (as if the user prompted a number between 1 and 4). Another test verifies that invalid user input is handled appropriately (when the user is prompting something different than any numbers between 1 and 4).

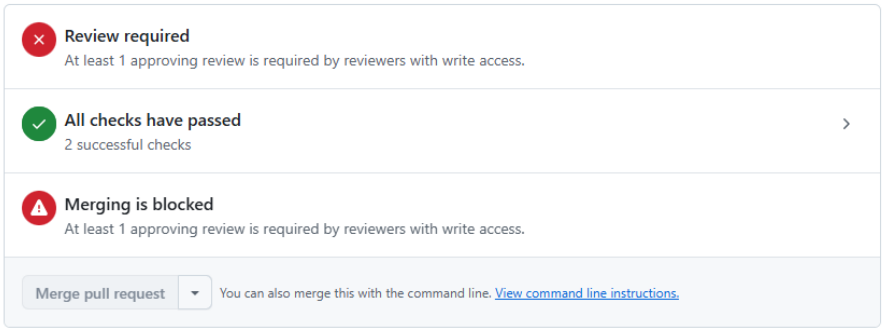
We also added three others unit tests: test_zero_score, which checks that a score of zero gives 0%; test_full_score, which checks that a perfect score gives 100%; and test_zero_questions, which verifies that zero questions return 0.0% without causing a division-by-zero error for edge case handling.

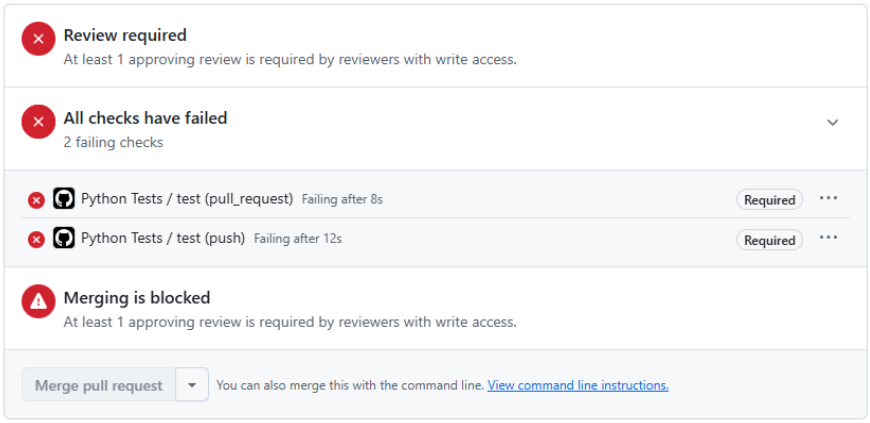
6. Continuous Integration Briefly explains when your GitHub Actions workflow runs, what its job/runner/steps do, one failed CI run, the cause of the failure, and how the group fixed it.

We configured a GitHub Actions workflow to automate testing. The workflow runs when changes are pushed and when pull requests are opened or updated. The workflow uses an Ubuntu runner, sets up Python application, installs the required testing dependencies, and runs the project's automated tests to detect problems before changes are merged into main.

During development, we intentionally added a failing test to demonstrate our branch protection rules. The test used to assert False to force the test suite to fail. GitHub Actions reported failing checks, and the pull request could not be merged because the required status checks had failed, and an approving review was missing.

We corrected it by choosing to not merge the intentionally failing test and kept it on a separate demonstration branch.

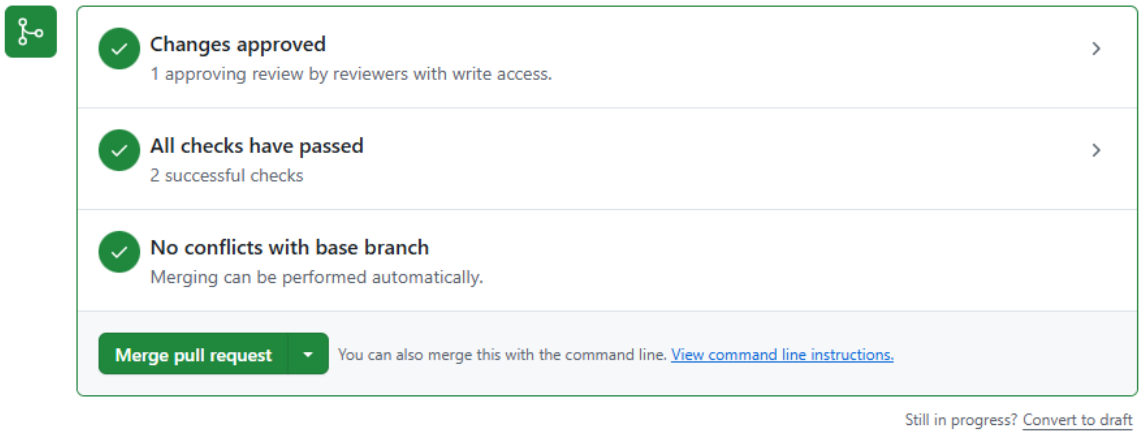




7. Ruleset & Protected main Describe the final main ruleset. Explain which review and CI requirements had to be satisfied before a Pull Request could be merged.

We configured a ruleset to protect the main branch by requiring all changes to be submitted through pull requests, at least one approving review from a reviewer with write access, and successful GitHub Actions status checks.

To demonstrate the review and CI requirements, we used a test pull request. When approval was missing, merging was blocked despite the successful test check. We also demonstrated that a failing automated check prevents merging. After restoring the test and obtaining approval, the pull request could be merged once both requirements were satisfied. (see the pictures above because they're also related)



8. Continuous Deployment Explain how the GitHub Pages site is published and describe the final change that was merged and then observed on the live site.

We configured GitHub Pages to publish our website from the /docs folder on the main branch. GitHub automatically builds and deploys the site when changes are merged into main. We submitted a final update to docs/index.html through a feature branch and pull request. After the pull request passed its required checks and was merged, we verified that the changes appeared on the live website via our screenshots.

Quiz Project

A simple Python quiz application.

Project Theme

Quiz: answering questions and checking the user's answers.

Team Members

- benlachhabrayan-glitch — Quiz feature and project coordination
- penouar — Additional quiz feature and testing

Replace these usernames or roles if your team distribution is different.

Implemented Features

- Python quiz application
- Quiz logic separated into a module
- Automated unit tests using pytest
- GitHub Actions workflow for automated testing
- Collaborative development using branches and pull requests

How to Run the Project

Install Python 3, then open a terminal in the project folder.

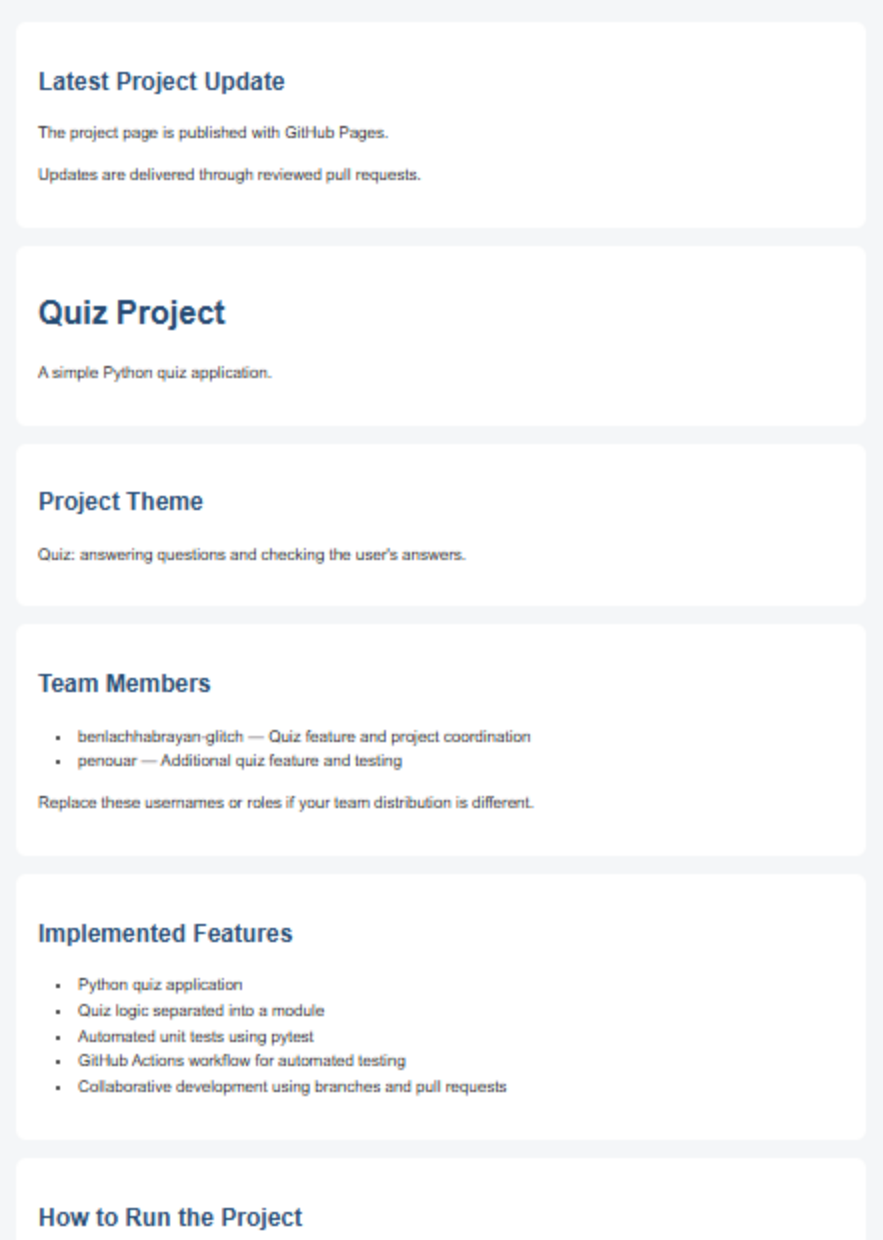
Install the dependencies:

```
python -m pip install -r requirements.txt
```

Run the quiz:

```
python quizz.py
```

Run the tests:



9. Contribution Summary For each member, list: feature(s), branch(es), Pull Request(s) authored, Pull Request(s) reviewed, and test contribution. Keep this concise.

Member 1 — Rayan Benlachhab

- Features: Quiz logic, automated tests, GitHub Actions, and GitHub Pages.
- Branches: feature/quiz-improvements and feature branches for demonstrations and website updates.

- Pull Requests authored: PR #1 and subsequent PRs for testing, branch protection demonstrations, and website updates.
- Pull Requests reviewed: Reviewed the collaborator's unit tests pull request, asked him to modify it and then accepted it after some changes were applied.
- Testing: Created and maintained unit tests, including normal and invalid-input cases.

Member 2 __Mohamed Amine Menouer

- Feature: Quiz-logic unit tests, edge-case testing, and GitHub collaboration workflow.
- Branch: feature/quiz-engine.
- Pull requests: Created the quiz-logic test pull request and participated in the README conflict exercise.
- Reviews: Reviewed the teammate's changes and participated in the request-changes cycle.
- Tests: Contributed tests for correct, incorrect, invalid, score-calculation, and zero-question cases.

10. Fair Use of AI Write one short paragraph following the assignment instructions. If different members used AI differently, identify those differences.

Rayan: I used AI to help guide me through the homework. Because we barely covered the use of GitHub with a collaborator and practiced it during the lab, it was a hard time understanding what to do by just looking at the requirements. The role of my AI agent was, as a tool, to tell me what to do in each situation and what does the homework really wants me to achieve. It also played a huge role when coding and prompting the exact code to communicate with Git Dash.

Amine: I used AI assistance to help understand Git and GitHub concepts, troubleshoot development and workflow issues, and improve the clarity and organization of the report. AI assistance was used as a support tool rather than a substitute for the required collaboration process. We reviewed the changes, tested the application, and used the actual repository and workflow results to complete the assignment.

11. Sources & Reflection List for useful sources used by the group. Then write 3–4 sentences on the hardest collaboration/automation problem you encountered and what your group would check first if a Pull Request, merge, test, or workflow behaves unexpectedly

We didn't use any external source during the group project, only relying on our bit of knowledge and the use of AI, the most difficult part was the fact that it was the first time using GitHub with collaborators, but also that working together remotely is really difficult in such situation, its achievable, but very time consuming, especially in this midterm period.

IMPORTANT: The account BetterCall911 is my backup account. It is tied to my AUI mail, but I tried to remove it from collaborators, but it don't show up even though trying to block it.

